VAIBHAV SINGH

Software Engineer | Full-Stack | Gen AI | Frontend/Backend Systems

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EXPERIENCE

Web Developer

NOV 2025 - FEB 2026

ECELL UIET KUK

Kurukshetra, India (On-site)

TECH STACK : HTML, CSS, JAVASCRIPT, REACT.JS, NODE.JS, MONGODB, EXPRESS.JS

WEBSITE

- Built and deployed ***React.js***-based web interfaces, improving load time by ***25%*** and increasing user engagement by ***30%.***
- Integrated ***REST APIs*** and optimized frontend performance, boosting data retrieval efficiency by ***40%*** during peak traffic.
- Delivered ***5+*** production features using ***Git-***based workflows, reducing manual effort by ***35%*** and improving system reliability.

SKILLS

- **Programming Languages:** C, C++, Python, Typescript, JavaScript, HTML, CSS.
- **Framework & Libraries:** React.Js, Node.Js, Next.Js, Express.Js.
- **Core Concepts:** Data Structures & Algorithms, DBMS, Operating Systems, Object-Oriented Programming, Digital Electronics, Advanced Mircoprocessor & Interfacing, Verilog.
- **Tools & Software:** Git & Github, Vercel, Postman, Canva, Figma, VS Code, Power BI, MATLAB, Webpack.
- **Machine Learning:** NumPy, Pandas, Plotly.
- **Database:** MySQL, MongoDB.

PROJECTS

Virtual Oscilloscope (Real-Time Signal Visualization Tool)

GITHUB LIVE

TECH STACK : Python, Numpy and Plotly.

- Developed a ***real-time signal*** visualization web app to simulate oscilloscope functionality with adjustable waveform parameters
- Implemented ***dynamic rendering of sine, square, and triangular waves*** with controls for frequency, amplitude, and time scaling
- Designed an interactive UI enabling voltage scaling and waveform analysis for better understanding of signals
- Ensured smooth performance using efficient state management and real-time updates

Biot–Savart Law Simulation (Magnetic Field Visualization Tool)

GITHUB LIVE

TECH STACK : Python, Numpy and Plotly.

- Built a **physics-based** simulation to **visualize magnetic field distribution** using the Biot–Savart Law.
- Implemented ***mathematical models to compute magnetic field intensity*** at various spatial points.
- Designed an interactive interface to demonstrate field variation with current and distance.
- Enabled ***real-time parameter manipulation*** for dynamic and intuitive visualization.
- Applied scientific computing and visualization techniques to simplify electromagnetic theory concepts.

Convo.ai (AI-chatbot for talking with documents)

GITHUB LIVE

TECH STACK : HTML, TalwindCSS, JAVASCRIPT, REACT.JS, Next.Js, OpenAI, Kinde Auth, Shadcn UI.

- Developed a ***real-time messaging web app*** using React.js, JavaScript, HTML, CSS, that enables intelligent ***conversations with PDF documents***.
- Implemented ***dynamic state management and asynchronous updates***, ensuring seamless message flow without page reloads
- Designed a responsive UI, improving accessibility and usability across devices
- Utilized ***Git & GitHub for version control***, streamlining collaboration and code management

EDUCATION

University Institute of Engineering & Technology

SEPT 2024 - JUNE 2028

Bachelor of Technology - Electronics & Communication Engineering

Kurukshetra, India

8.2 CGPA

CERTIFICATIONS & ACHIEVEMENTS

- Tech Team Co-Lead, College Entrepreneur Club - Led a team of 8+ developers to build and maintain web platforms, increasing event participation by 40%.
- Completed HackerRank Skills Test: Python Badge CSS Badge React.Js Badge
- Completed Neural Networks & Deep Learning Course Work from NIT KURUKSHETRA
- Completed Microsoft: Generative AI career essentials course.
- Achieved Atlassian IT Service Management (ITSM) Professional Certificate.